

L2: Random Walks

In some cases, we do not have a complete *map* of the network. Instead, we only know the node that we currently reside at, and the connections of that node. When that is the case, it is often useful to perform a **random walk** in which we move randomly between network nodes.

The simplest example of a random walk is to imagine a **walker** that is stationed at node v . The walker then randomly selects a neighbor of v and moves to that neighbor. If the network is unweighted, the walker will transition along each edge with a probability of $\frac{1}{k}$ where k is the number of outgoing edges for that particular node. If a network is weighted, the transition probabilities are functions of the edge weights. These transition probabilities can be represented with a matrix P in which the (i, j) element is the probability that the walker moves from node i to node j .

If the walker continues randomly moving from neighbor to neighbor and recording the number of times it transitions along a particular edge, a probability distribution of finding the walker on a particular node at any given time emerges. This distribution is known as the **stationary distribution**.

So how can we calculate this distribution using our transition matrix P ? There are two ways to approach this. For the first, we need a vector that represents the walker probability for each node at a time t . Let's call this vector q_t . Often we are given the probability values for the initial state of each node, but the stationary distribution is not dependent on the initial state probabilities. This means we can assign each node any initial probability so long as they all add up to 1. Next, we describe each iteration of the random walk by the equation: $q_{t+1} = P^T q_t$, where P^T is the transpose of the transition matrix. For each iteration of t , the i_{th} element of the resulting vector q_{t+1} is nothing but the probability of i being the current node calculated as the probability incoming edge (j, i) was taken, times the probability that the walker was at previous node j . We can express this as

$$Prob(current_{node} = i) = \sum_{j=1}^N Prob(edge(prev_{node} = j, current_{node} = i)) \times Prob(prev_{node} = j),$$

where N is the total number of nodes in the graph. As t increases, the probability values of q_t will converge asymptotically. Note that the sum of the elements of q_t will remain equal to 1 for any time t .

Even though this method takes several iterations of time to find the distribution, note that the distribution itself does not change over time. Exploiting this fact leads us to our second method as follows.

Let q be the stationary distribution expressed as a column vector. It satisfies the relationship $P^T q = q$ for transition matrix P . Recall from linear algebra that a transition matrix T has an eigenvector v if $Tv = \lambda v$ for an eigenvalue λ . From this, we can see that the eigenvectors of P^T are the stationary distribution expressed as column vectors where the eigenvalue $\lambda = 1$.

An important result of this is that, in undirected and connected networks, a stationary distribution always exists. It is not, however, necessarily unique. See also the first "food-for-thought" question below.



Food for Thought

1) Show that in undirected and connected networks in which the elements of the matrix $\mathbf{P}$ are strictly positive (and so there is at least a small probability of transitioning from every node to every other node), the steady-state probability vector $\mathbf{q}$ is unique and it is the leading eigenvector of the transition matrix $\mathbf{P}^T$.

Hint: the largest eigenvalue of $\mathbf{P}^T$ is equal to 1. Why?

2) What can go wrong with the stationary distribution equation in directed networks?